

Proof. Fix $x \in C_k$ and let $(y, z) \in E^s$ with $y \in C_k, z \in C_\ell$. The event $\{\bar{\tau}_{C_\ell}^\varepsilon < \bar{\tau}_x^\varepsilon\}$ occurs if y is hit before returning to x and the edge to z is immediately taken, so that

$$\mathbb{P}_x(\bar{\tau}_{C_\ell}^\varepsilon < \bar{\tau}_x^\varepsilon) \geq p^\varepsilon(z|y)\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_x^\varepsilon) \geq \frac{\nu c \varepsilon}{2 \log M}.$$

Hence by Proposition C.10 we obtain

$$q_\star^\varepsilon(C_\ell|C_k) = \mu_k(x)\mathbb{P}_x(\bar{\tau}_{C_\ell}^\varepsilon < \bar{\tau}_x^\varepsilon) \left(1 + \tilde{O}\left(\frac{1}{\log M}\right)\right) \geq \Omega\left(\frac{\varepsilon}{M \log M}\right).$$

□

C.3 Hitting Time Analysis

To prove Theorem 3.2, we first derive the expected escape time of a single cluster.

Lemma C.12. *For all $k \in [K]$ and $x \in C_k$, it holds that*

$$\mathbb{E}_x[\tau_{C_k^c}^\varepsilon] = \tilde{\Theta}\left(\frac{M}{\varepsilon}\right).$$

Proof. Recall from Lemma B.9 that the mixing time of $\tilde{X}^{k,\varepsilon}$ is $t_{\text{mix}} = O(\log M)$. Also denote the set of states in C_k with outbound edges as $D_k := \{x \in C_k : \exists y \notin C_k, (x, y) \in E_s\}$, so that $1 \leq |D_k| \leq n_{\text{out}}$. By Assumption 2. Since $\tau_{C_k^c}^\varepsilon > m$ implies that $\tilde{X}_t^{k,\varepsilon} = X_t^\varepsilon$ for $t \leq m$ and that a sparse edge was not taken at each state of the skipped subchain $\tilde{X}_{t_{\text{mix}}t}^{k,\varepsilon}$ up to $t = \lfloor m/t_{\text{mix}} \rfloor$, it follows that

$$\begin{aligned} \mathbb{P}_x(\tau_{C_k^c}^\varepsilon > m) &\leq \sup \prod_{t=1}^{\lfloor m/t_{\text{mix}} \rfloor} \mathbb{P}_{X_{t_{\text{mix}}(t-1)+1}^\varepsilon}(X_{t_{\text{mix}}t+1}^\varepsilon \in C_k) \\ &\leq \sup \prod_{t=1}^{\lfloor m/t_{\text{mix}} \rfloor} \left(1 - \mathbb{P}_{X_{t_{\text{mix}}(t-1)+1}^\varepsilon}(\tilde{X}_{t_{\text{mix}}t}^{k,\varepsilon} \in D_k) \cdot \mathbb{P}_{\tilde{X}_{t_{\text{mix}}t}^{k,\varepsilon}}(X_{t_{\text{mix}}t+1}^{k,\varepsilon} \notin C_k)\right) \\ &\leq \prod_{t=1}^{\lfloor m/t_{\text{mix}} \rfloor} \left(1 - \frac{\rho|D_k|}{2M} \cdot c\varepsilon\right) \\ &\leq \exp\left(-\frac{\rho c \varepsilon m}{2M t_{\text{mix}}}\right). \end{aligned}$$

This yields the upper bound

$$\mathbb{E}_x[\tau_{C_k^c}^\varepsilon] = \sum_{m=0}^{\infty} \mathbb{P}_x(\tau_{C_k^c}^\varepsilon > m) \leq \left(1 - \exp\left(-\frac{\rho c \varepsilon}{2M t_{\text{mix}}}\right)\right)^{-1} \leq O\left(\frac{M \log M}{\varepsilon}\right).$$

For the lower bound, consider the partition of $(\tilde{X}_t^{k,\varepsilon})_{t \geq 0}$ into the union of skipped and shifted subchains $(\tilde{X}_{t_{\text{mix}}t+u}^{k,\varepsilon})_{t \geq 0}$ for $0 \leq u < t_{\text{mix}}$. Suppose that $m \geq 2t_{\text{mix}}$, so each subchain has length at least 2, and all transition probabilities of each subchain is $\Theta(1/M)$ by Assumption 1. Since not taking a sparse edge at each step of all subchains implies $\tau_{C_k^c}^\varepsilon > m$,

$$\begin{aligned} & \mathbb{P}_x(\tau_{C_k^c}^\varepsilon > m) \\ & \geq \inf \prod_{u=0}^{t_{\text{mix}}-1} \prod_{t=1}^{\lfloor m/t_{\text{mix}} \rfloor} \left(1 - \mathbb{P}_{\tilde{X}_{t_{\text{mix}}(t-1)+u+1}^{k,\varepsilon}}(\tilde{X}_{t_{\text{mix}}t+u}^{k,\varepsilon} \in D_k) \cdot \mathbb{P}_{\tilde{X}_{t_{\text{mix}}t+u}^{k,\varepsilon}}(X_{t_{\text{mix}}t+u+1}^\varepsilon \notin C_k) \right) \\ & \geq \prod_{u=0}^{t_{\text{mix}}-1} \prod_{t=1}^{\lfloor m/t_{\text{mix}} \rfloor} \left(1 - \Theta\left(\frac{|D_k|}{M} \cdot d_{\text{out}}\varepsilon\right) \right) \\ & \geq \left(1 - \Theta\left(\frac{\varepsilon}{M}\right) \right)^{m-t_{\text{mix}}}. \end{aligned}$$

Note that while the dependency on t_{mix} does not explicitly appear in the bound, t_{mix} still needs to be small enough to argue that the states of each subchain for $t \geq 1$ exist and are sufficiently mixed. Hence it follows that

$$\mathbb{E}_x[\tau_{C_k^c}^\varepsilon] = \sum_{m=0}^{\infty} \mathbb{P}_x(\tau_{C_k^c}^\varepsilon > m) \geq \Omega\left(\frac{M}{\varepsilon}\right) - 2t_{\text{mix}} \geq \Omega\left(\frac{M}{\varepsilon}\right),$$

which concludes the statement. $\square$

Theorem C.13 (Theorem 3.2 restated). *Under Assumptions 1-4, it holds for all $\varepsilon \leq \varepsilon_{\text{max}} := \Theta(M^{-1}(\log M)^{-4})$ that*

$$\mathbb{E}_{(X_{\text{in}}, X_{\text{out}}) \sim \mathcal{D}}[\mathbb{E}_{X_{\text{in}}}[\tau_{X_{\text{out}}}^\varepsilon]] = \tilde{\Theta}\left(\frac{KM}{\varepsilon}\right).$$

Proof. Suppose $X_{\text{in}} \in C_k, X_{\text{out}} \in C_\ell$ with $k \neq \ell$. For the upper bound, by (7) it holds that

$$\mathbb{E}_{X_{\text{in}}}[\tau_{X_{\text{out}}}^\varepsilon] = \mathbb{E}_{X_{\text{in}}}[\bar{\tau}_{X_{\text{out}}}^\varepsilon] \leq \mathbb{E}_{X_{\text{in}}}[\bar{\tau}_{X_{\text{out}}}^\varepsilon] + \mathbb{E}_{X_{\text{out}}}[\bar{\tau}_{X_{\text{in}}}^\varepsilon] = \frac{1}{\pi^\varepsilon(X_{\text{in}})\mathbb{P}_{X_{\text{in}}}(\bar{\tau}_{X_{\text{out}}}^\varepsilon < \bar{\tau}_{X_{\text{in}}}^\varepsilon)}.$$

Combining (14), Corollary C.7 and Proposition C.10 yields

$$\begin{aligned} \pi^\varepsilon(X_{\text{in}})\mathbb{P}_{X_{\text{in}}}(\bar{\tau}_{X_{\text{out}}}^\varepsilon < \bar{\tau}_{X_{\text{in}}}^\varepsilon) &= \pi^\varepsilon(C_k) \cdot \frac{\pi_k^\varepsilon(X_{\text{in}})}{\mu_k(X_{\text{in}})} \cdot \mu_k(X_{\text{in}})\mathbb{P}_{X_{\text{in}}}(\bar{\tau}_{X_{\text{out}}}^\varepsilon < \bar{\tau}_{X_{\text{in}}}^\varepsilon) \\ &= \pi^\varepsilon(C_k)q_\star^\varepsilon(C_\ell|C_k) \cdot \left(1 + \tilde{O}\left(\frac{1}{\log M}\right) \right) \\ &= O\left(\frac{\varepsilon}{KM}\right), \end{aligned}$$

where the last line follows from Corollary C.9 and Assumption 4.

For the lower bound, define the sequence of increasing stopping times $(\sigma_n)_{n \geq 0}$ as

$$\sigma_0 := 0, \quad \sigma_n := \min\{t > \sigma_{n-1} : (X_{t-1}^\varepsilon, X_t^\varepsilon) \in E_s\}.$$

Then defining the minimum number of cluster transitions to reach X_{out} as

$$N = N(X_{\text{in}}, X_{\text{out}}) := \min\{|X_{0:T} \cap E_s| : X_0 = X_{\text{in}}, X_T = X_{\text{out}}, (X_{t-1}, X_t) \in E \forall t\},$$

it holds that $\tau_{X_{\text{out}}}^\varepsilon \geq \sigma_N$. Moreover denoting the cluster containing $X_{\sigma_{t-1}}^\varepsilon$ as $C[t]$, by Lemma C.12 we have

$$\mathbb{E}_{X_{\text{in}}}[\sigma_N] = \sum_{t=1}^N \mathbb{E}_{X_{\sigma_{t-1}}^\varepsilon}[\tau_{C[t]}^\varepsilon] \geq \tilde{\Theta}\left(\frac{MN}{\varepsilon}\right),$$

and hence

$$\mathbb{E}_{(X_{\text{in}}, X_{\text{out}}) \sim \mathcal{D}}[\mathbb{E}_{X_{\text{in}}}[\tau_{X_{\text{out}}}^\varepsilon]] \geq \tilde{\Theta}\left(\frac{KM}{\varepsilon}\right)$$

since $\mathbb{E}[N] = \Omega(K)$ by Assumption 3. □

D Proofs for Optimization Dynamics

D.1 Analysis of Pretraining Dynamics

Theorem D.1. *Let $X_0 \sim \text{Unif}(S)$ or $X_0 \sim \pi^\varepsilon$ and $X_1 \sim p^\varepsilon(\cdot|X_0)$ be random samples from the Markov chain X^ε . Then:*

- (1) *The sequence of gradient descent iterates $(\mathbf{W}^{(t)})_{t \geq 0}$ for cross-entropy loss*

$$L_{\text{pre}}(\mathbf{W}) = \mathbb{E}_{X_0, X_1}[-\log \hat{p}_{\mathbf{W}}(X_1|X_0)]$$

with initialization $\mathbf{W}^{(0)} = \mathbf{0}$ and suitable learning rate converges with respect to the learned transition probabilities as

$$\sup_{1 \leq i, j \leq S} |\hat{p}_{ij}^{(T)} - p_{ij}^\varepsilon| = O\left(\sqrt{\frac{KM^2}{T}} \log \frac{KT}{M\varepsilon}\right). \quad (17)$$

- (2) *After $T_1 = \tilde{O}(KM^2\varepsilon^{-2})$ steps, by setting $w_{ij} \leftarrow -\infty$ if $\hat{p}_{ij}^{(T_1)}$ is below a threshold $c_{\text{thres}}\varepsilon$ it holds for the resulting model $\hat{p}$ that $\hat{p}_{ij} = 0$ iff $p_{ij}^\varepsilon = 0$ and*

$$p_{ij}^\varepsilon - o(\varepsilon) \leq \hat{p}_{ij} \leq p_{ij}^\varepsilon + o(1) \quad (18)$$

holds uniformly for all j such that $p_{ij}^\varepsilon \neq 0$.